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Micro-environment apparatus and method for air sensor

Abstract

A remote test apparatus is configured to simulate an environment of at least one remotely located room and configured to receive at least one air sample from the at least one remotely located room. The apparatus includes a sealed enclosure forming an interior cavity and including an inlet in connection with the at least one remotely located room and an outlet in connection with a sample collection unit. The sample collection unit is configured to communicate the at least one air sample from the remotely located room at a first flow rate. The apparatus further includes at least one sensor disposed in the interior cavity and at least one air transfer unit configured to transfer test air from the interior cavity of the sealed enclosure to the at least one sensor at a second flow rate.

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References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5184501	12/1992	Lewis et al.	N/A	N/A
5292280	12/1993	Janu et al.	N/A	N/A
5293771	12/1993	Ridenour	N/A	N/A
5357781	12/1993	Tikijian	N/A	N/A
5394934	12/1994	Rein et al.	N/A	N/A
5464369	12/1994	Federspiel	N/A	N/A
5550752	12/1995	Federspiel	N/A	N/A
5976010	12/1998	Reese et al.	N/A	N/A
6125710	12/1999	Sharp	N/A	N/A
6241950	12/2000	Veelenturf et al.	N/A	N/A
6425297	12/2001	Sharp	N/A	N/A
7302313	12/2006	Sharp et al.	N/A	N/A
7389158	12/2007	Desrochers et al.	N/A	N/A
7434413	12/2007	Wruck	N/A	N/A
8066558	12/2010	Thomle et al.	N/A	N/A
8147302	12/2011	Desrochers et al.	N/A	N/A
9109981	12/2014	Sharp	N/A	N/A
2013/0260668	12/2012	Stakutis et al.	N/A	N/A
2015/0253165	12/2014	Ajay et al.	N/A	N/A
2015/0323427	12/2014	Sharp	N/A	N/A
2017/0158020	12/2016	Park	N/A	G01N 21/534
2018/0363939	12/2017	McCormick	N/A	G01N 1/24
2020/0264155	12/2019	Alves Fortunato	N/A	G01N 33/22

FOREIGN PATENT DOCUMENTS

Patent No.	Application Date	Country	CPC
201589957	12/2009	CN	N/A

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims priority under 35 U.S.C. § 119(e) and the benefit of U.S. Provisional Application No. 63/016,023 entitled MICRO-ENVIRONMENT APPARATUS AND METHOD FOR AIR SENSOR, filed on Apr. 27, 2020, by McManus et al., the entire disclosure of which is incorporated herein by reference.

TECHNOLOGICAL FIELD

(1) The present disclosure generally relates to air monitoring systems and, more particularly, relates to air monitoring systems utilizing multipoint air sampling.

BACKGROUND OF THE DISCLOSURE

(2) Building air supply and control systems are becoming increasingly important due to increases in energy cost. In various building ventilation systems (e.g. HVAC systems), it may be beneficial to improve data acquisition for each room of the building. The disclosure provides for a variety of improvements in air sampling and air quality control for building ventilation and control systems.

BRIEF SUMMARY

(3) According to one aspect of the present disclosure, a remote test apparatus is configured to simulate an environmental condition of at least one remotely located room. In operation, the apparatus is configured to receive at least one air sample from the at least one remotely located room. The apparatus comprises a sealed enclosure forming an interior cavity and comprising an inlet in connection with the at least one remotely located room and an outlet in connection with a sample collection unit. The sample collection unit is configured to draw the at least one air sample from the remotely located room at a first flow rate. The apparatus further comprises at least one sensor disposed in the interior cavity and at least one air transfer unit configured to transfer test air from the interior cavity of the sealed enclosure to the at least one sensor at a second flow rate.

(4) According to another aspect of the disclosure, a method for simulating an environmental condition of at least one remotely located room is disclosed. The method includes supplying at least one air sample at a first flow rate from the at least one remotely located room to an interior cavity of a sealed enclosure. The method further includes sampling the at least one air sample within the interior cavity at a second flow rate different than the first flow rate and detecting the environmental condition in the at least one air sample at the second flow rate within the interior cavity. A test environment within the interior cavity simulates the environmental condition of the remotely located room.

(5) In yet another aspect of the disclosure, a remote test apparatus is configured to simulate an environment of at least one remotely located room and configured to receive at least one air sample from the at least one remotely located room. The apparatus includes a sealed enclosure forming an interior cavity formed by a sealed boundary wall. The sealed enclosure includes an inlet in connection with the at least one remotely located room and an outlet in connection with a sample collection unit. The sample collection unit is configured to transfer the at least one air sample from the remotely located room at a first flow rate. The test apparatus also includes at least one sensor disposed in the interior cavity and a pressure sensor that detects a pressure differential between the interior cavity and a local environment proximate the sealed enclosure. A controller is configured to detect a system error in response to changes in the pressure differential.

(6) These and other aspects, objects, and features of the present disclosure will be understood and appreciated by those skilled in the art upon studying the following specification, claims, and appended drawings. It will also be understood that features of each embodiment disclosed herein may be used in conjunction with, or as a replacement for, features of the other embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) In the drawings:
- (2) FIG. 1 is a block diagram of a building demonstrating a plurality of rooms incorporating an air sampling system;
- (3) FIG. 2 is a block diagram of a building having an air sampling system;
- (4) FIG. 3 is a schematic diagram of the air sampling system;
- (5) FIG. 4 is a detailed schematic diagram of the sample return unit of the air sampling system;
- (6) FIG. 5 is a cross-sectional diagram of a remote test apparatus configured to simulate ambient conditions of a room;
- (7) FIG. 6 is a side cross-sectional view of a remote test apparatus demonstrating a plurality of sensors in a micro-environment;
- (8) FIG. 7 is a top cross-sectional view of a remote test apparatus demonstrating a plurality of sensors in a micro-environment; and
- (9) FIG. 8 is a flow chart demonstrating a method for controlling a remote test apparatus to detect simulated environmental conditions for a room in connection with an air sampling system in accordance with the disclosure.

DETAILED DESCRIPTION OF EMBODIMENTS

- (10) The present illustrated embodiments reside primarily in combinations of method steps and apparatus components related to an air monitoring device. Accordingly, the apparatus components and method steps have been represented, where appropriate, by conventional symbols in the drawings, showing only those specific details that are pertinent to understanding the embodiments of the present disclosure so as not to obscure the disclosure with details that will be readily apparent to those of ordinary skill in the art having the benefit of the description herein. Further, like numerals in the description and drawings represent like elements.
- (11) In this document, relational terms, such as first and second, top and bottom, and the like, are used solely to distinguish one entity or action from another entity or action, without necessarily requiring or implying any actual such relationship or order between such entities or actions. The terms “comprises,” “comprising,” or any other variation thereof, are intended to cover a non-exclusive inclusion, such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements but may include other elements not expressly listed or inherent to such process, method, article, or apparatus. An element preceded by “comprises . . . a” does not, without more constraints, preclude the existence of additional identical elements in the process, method, article, or apparatus that comprises the element.
- (12) Modern heating, ventilation, and air conditioning (HVAC) systems may provide for independent control of various regions or rooms within a building. In this way, climate control and air quality may be monitored and controlled for each region or room. In addition to the benefits of improved comfort and air quality, systems may also be configured to reduce energy consumption. In some cases, conserving energy may include determining occupancy of each room or region. In response to the occupancy or level of occupancy, the HVAC setting may be optimized to ensure that the air quality is maintained. However, there are a variety of challenges that may be associated with implementing such systems on a large scale while maintaining economical HVAC control solutions. The following detailed description provides for a system and control method for an air

sampling system that is both effective and economical.

(13) Referring to FIGS. 1 and 2, block diagrams of a building **10** having an air sampling system **100** is shown. The air sampling system **100** may be configured to selectively measure at least one air quality characteristic from a plurality of areas of the building **10**. For example, the system **100** may be configured to monitor various volumetric regions of a single room and/or monitor the conditions of each of a plurality of rooms **102** (e.g. rooms 11-14, 21-24, 51-54, etc.). The plurality of rooms **102** or areas of the building **10** may correspond to regions or volumetric spaces within a commercial building, residential building, multi-purpose building, etc. The disclosure may provide for various novel aspects configured to limit the installation and operating cost of the system **100**, and may also provide for improved operational performance.

(14) In an exemplary embodiment, the system **100** may comprise an air sampling device **104**. The air sampling device **104** may correspond to a scalable monitoring device configured to sample the air from each of the plurality of rooms **102** via an air flow controller **106**. The air flow controller **106** may comprise a header configured to receive multiple inputs corresponding to each of the plurality of rooms **102**. The air flow controller **106** may be configured to selectively direct a room air sample from a selected room **108** (e.g. room 12) to one or more sensors **110**. While drawing the room air sample from the selected room **108**, the air sampling device **104** may be configured to draw air or communicate air from the idle rooms **112** of the plurality of rooms **102**. In this configuration, the sampling device may be operable to monitor a condition of the air in each of the plurality of rooms **102** with only limited and consistent delay to gather a current air sample. As will be discussed further herein, the delay of the air sampling device **104** to update a sample for each selected room may be consistent regardless of a distance of the selected room **108** from the sensor **110** of the sampling device **104**.

(15) The system **100** may further comprise an air transport motivator or sample return unit **114**. The sample return unit **114** may correspond to an energy efficient air transport unit configured to function without a need for direct electrical power. In such embodiments, the sample return unit **114** may be configured to generate suction to draw the room air samples from each of the plurality of rooms **102** via a bulk supply fan of an existing heating, ventilation, and air conditioning (HVAC) system **116** of the building **10**. For example, the sample return unit **114** may correspond to a Venturi vacuum device configured to utilize differential pressure derived from the bulk supply fan of the HVAC system **116**. Further details of the sample return unit **114** are discussed in reference to FIG. 3. The HVAC system **116** may be configured to provide heating, ventilation, and air conditioning to each of the rooms **102**. In this configuration, the sampling return unit **114** may be configured to function without a direct supply of electrical energy.

(16) Though the sample return unit **114** is described in the exemplary embodiment as a Venturi vacuum device, in some embodiments, the sample return unit **114** may recover the room air samples utilizing conventional vacuum pumps or fans. For example, in some embodiments, the sample return unit **114** may utilize suction created via a reciprocating pump, screw or turbine compressor, or various devices configured to displace air in response to receiving electrical current. Such devices may be utilized to provide vacuum pressure to return the room air samples. However, these devices may be noisy, inefficient, and require more maintenance than the Venturi vacuum device disclosed herein.

(17) In the exemplary embodiment utilizing the Venturi vacuum device, the only energy usage of the sample return unit **114** may be a small parasitic energy drawn from the bulk supply fan of the HVAC system **116**. For example, the bulk supply fan may consume a small and potentially negligible increased power due to the energy usage of the sample return unit **114**. The change in power usage may particularly be negligible when the air sampling system **100** is implemented in large buildings having multiple rooms and high volume HVAC systems **116**. In this way, the air sampling system **100** may improve efficiency while limiting components and related maintenance that may be required when utilizing active devices to provide vacuum pressure to return the room

air samples. As disclosed, the sample return unit **114** may utilize the existing fluid pressure from the HVAC system **116** to power the sample return unit **114** via the Venturi effect to improve efficiency and limit maintenance.

(18) In operation, the sample return unit **114** may draw room air samples from each of the plurality of rooms **102** perpetually during operation of the bulk supply fan of the HVAC system **116**. For example, the controller **118** of the air sampling system **100** may be configured to control the air flow controller **106** to direct a room air sample of a selected room **108** to a quality sensor of the at least one sensor **110** to measure a condition or quality parameter of the selected room **108**. The air quality sensor is identified as element **140** in FIG. 3 and is discussed further in reference to FIG. 3. Additionally, the air flow controller **106** may be configured to retrieve and purge room air samples **132** from each of the idle rooms **112** (e.g. rooms 11, 13, 14, 21-24, 51-54, etc.).

(19) By continuously drawing the room air samples from each of the idle rooms **112** and the selected room **108**, the air sampling system **100** may continuously supply updated air samples to the air flow controller **106**. In this way, the system **100** may supply air samples from each of the rooms **102** to a central location of the air flow controller **106** and selectively direct the air sample from the selected room **108** to the air quality sensor **140**. Accordingly, the system **100** may avoid delays in updating the air samples from each of the idle rooms **112** that may otherwise be required to purge an associated length of each of the supply lines **120** or more specifically, each of a plurality of room air sample supply lines **120a** from each of the rooms **102** to the air flow controller **106**. By continuously supplying updated air samples from each of the rooms **102** containing air that is representative of a current air quality, the sampling system **100** is operable to test each of the rooms **102** without significant delay that may otherwise be required to purge air samples.

Accordingly, the air supplied from the air flow controller **106** to the at least one sensor **110** may only need to be purged from the air in the sensor supply line **120b** rather than each of the air sample supply lines **120a** when changing the selected room **108** among the plurality of rooms **102**.

(20) In operation, the controller **118** may measure the air quality of one or more of the rooms **102** by cycling the selected room **108** to supply an air sample from one or more of the rooms **102**. For example, during such an operation, the controller **118** may control the air flow controller **106** and the at least one sensor **110** to measure the air quality of room 12. Following the measurement of the air quality in room 12, the controller **118** may generate a control signal configured to change the air flow controller **106** to select room 22 as the selected room **108**. In this configuration, the idle rooms **112** may include rooms 11-14, 21, 23, 24, 51-54; the air samples from these rooms may be purged while the selected room **108** is designated as room 22.

(21) The room air sample from room 22 may be directed by the air flow controller **106**, such that the air entering the at least one sensor **110** is from room 22. The controller **118** may then allow a predetermined period of time for the room air sample from room 12 to purge from a sensor supply line **120b** and to allow time for a change in the sampled air to become stable in the sensor **110**. Once the predetermined period of time has lapsed, the controller **118** may monitor the at least one sensor **110** to identify a condition or quality parameter of the room air sample for room 22. If the room air samples from each of the rooms **102** were not delivered to the air flow controller **106** throughout operation, the purge time may increase dramatically and vary for each of the rooms **102** due to variations in the lengths of the air sample supply lines **120a**. Such variations and increases in purge time may result in delayed correction of air quality in each of the rooms **102** as well as increased maintenance and operational cost for the sampling system **100**.

(22) The controller **118** of the air sampling system **100** may continue to control the air flow controller **106** to test the condition or an air quality parameter of each of the rooms **102** in any sequence. The controller **118** may be configured to monitor each of the room air samples for each of the rooms **102** for a monitoring period, which may be independently specified for one or more of the rooms **102**. Additionally, the air sampling system **100** may be configured to monitor the rooms **102** sequentially or in any order. For example, the controller **118** may be configured to control the

air flow controller **106** to monitor a frequently occupied room with a greater frequency than other rooms of the plurality of rooms **102**.

(23) As discussed previously, the continuous delivery of the room air samples for each of the rooms **102** by the air flow controller **106** may limit a purging time required to gather a current or updated sample from the selected room **108**. Additionally, the continuous delivery of the room air samples may limit a flow rate necessary for the sample return unit **114**. In this way, the air sampling system **100** can maintain a low cycle time allowing the controller **118** to monitor the condition or an air quality parameter of each of the rooms **102** over a limited cycle time. The cycle time may correspond to a time required for the controller **118** to measure the condition or an air quality parameter of each of the rooms **102** and may also be referred to as a refresh rate.

(24) As further discussed in reference to FIGS. 5, 6, 7, and 8; in some implementations, the system **100** may comprise a remote test environment **122**, which may utilize a sealed enclosure **124** to create a micro-environment **126** for each of the selected rooms **108**. The remote test environment **122** may be formed within the sealed enclosure **124**, which may be housed within the sampling device **104**. In operation, the air flow controller **106** may continuously cycle sampled air from each of the rooms **102** and selectively deliver the air from the selected room **108** into the sealed enclosure **124** such that the micro-environment **126** is generated within an interior cavity **128** of the sealed enclosure **124**. In this way, the system **100** may provide for a simulated environment with representative air quality conditions to that of the selected room **108** and each of the rooms **102** in a remote location separate from each of the rooms **102** within the sampling device **104**.

(25) Within the remote test environment **122** formed by the sealed enclosure **124**, at least one of the sensors **110** may be housed such that the air flow controlled by the air flow controller **106** may be delivered from the selected room **108** into the interior cavity **128**. In this way, the system may generate similar air conditions to those of the selected room **108** within the sealed enclosure **124**, such that the one or more sensors **110** disposed therein may be operable to sample and test the micro-environment **126** and thereby simulate a test of the ambient air conditions within the selected room **108**. The operation of the remote test environment **122** may be cycled for each of the rooms **102** in connection with the sampling device **104** without appreciable lag or delay as a result of the continuous delivery of the room air samples from each of the rooms **102**. In this way, the system **100** may provide for advanced testing of simulated ambient air conditions in each of the rooms **102** while only requiring a limited vacuum pressure and corresponding flow rate of the room air samples supplied to the sampling device **104** by the flow controller **106**.

(26) Systems that do not utilize continuous delivery of the room air samples may require a significant increase in cycle time to monitor the rooms **102** or an increased flow rate for an air sample return unit. For example, the cycle time may increase due to increased purge time required to receive air sample data representative of the current air quality for a selected room **108**. The flow rate of the air samples recovered from each room may be increased to limit the purge time, but such operation may require an air sample return device having increased power consumption, which may reduce operational efficiency and increase the noise of operation. Accordingly, the systems and methods disclosed provide for the air sampling system **100** to operate with improved efficiency while maintaining a responsive system having a short cycle time.

(27) Referring now to FIG. 3, a schematic diagram of the air sampling system **100** is shown. As illustrated in FIG. 3, the air flow controller **106**, the at least one sensor **110**, and the sample return unit **114** are demonstrated in further detail. The air flow controller **106** may comprise a multi-input header **130** configured to receive each of a plurality of room air samples **132** from the plurality of rooms **102**. The controller **118** may be configured to output a control signal to control the air flow controller **106** identifying a room of the plurality of rooms **102** to be measured for an air quality parameter or air condition. In response to receiving the signal, the air flow controller **106** may control one or more valves to direct the room air sample from the selected room **108** to the at least one sensor **110**. In this way, the room air sample from the selected room **108** may be directed to the

at least one sensor **110**.

(28) During operation, the room air samples **132** from each of the plurality of rooms **102** may be consistently delivered to the air flow controller **106** via the air sample supply lines **120a**, each of which is in fluid communication with the air in one of the plurality of rooms **102**. The room air samples **132** may be delivered at a rate of a volumetric displacement of air induced by a pressure differential generated in the sample return unit **114**. The room air sample from the selected room **108** may be delivered to the at least one sensor **110**. From the at least one sensor **110**, the air may be drawn into the sample return unit **114** via a coupler **134**. Concurrently to the delivery of the room air sample from the selected room **108**, the room air samples from the idle rooms **112** may be drawn through the air flow controller **106** to the sample return unit **114** via the coupler **134**. In this configuration, the room air samples **132** from each of the plurality of rooms **102** may be consistently delivered to the air flow controller **106** to provide updated room air samples for each of the plurality of rooms **102** throughout operation.

(29) The at least one sensor **110** may correspond to a variety of types of sensors configured to measure one or more properties or air quality parameters of the room air samples **132**. In some embodiments, the at least one sensor **110** may correspond to one or more air quality sensors **140** configured to detect a condition or change in condition in each of the room air samples **132**. The air quality sensor **140** may be configured to communicate one or more measurements or other data to the controller **118**. The controller **118** may process the one or more measurements or other data to identify air quality conditions or metrics for each of the plurality of rooms **102**.

(30) The measurements communicated to the controller **118** may be utilized to determine an appropriate action to improve the air quality for a specific room of the plurality of rooms **102**. In this way, the controller **118** may identify a condition or change in condition in one or more air quality parameters. Based on the identified condition or change, the controller **118** may communicate a ventilation control signal configured to induce a building management system (BMS) **142** to take corrective action. In response to the ventilation control signal, the BMS **142** may control the HVAC system **116** to provide ventilation, heat, and/or cooled air to a specific room or group of rooms of the plurality of rooms **102**.

(31) In an exemplary embodiment, the air quality sensor **140** may correspond to a carbon dioxide (CO.sub.2) sensor. The CO.sub.2 sensor may be configured to measure a CO.sub.2 level of the air in the selected room **108**. In such an embodiment, the air quality sensor **140** (e.g. the CO.sub.2 sensor) may be configured to communicate the CO.sub.2 level of the selected room **108** to the controller **118**. Based on a change or measurement of the CO.sub.2 level, the controller **118** may communicate a ventilation control signal to the BMS **142**. Based on the ventilation control signal, the BMS **142** may control the HVAC system **116** to provide ventilation, heating, or cooling to the selected room **108** (e.g. room 12). In this way, the system **100** may be configured to monitor the air quality of each of the rooms **102** and communicate the CO.sub.2 level to the BMS **142**.

Accordingly, the system **100** may be utilized to determine if the ventilation supplied by the HVAC system **116** is sufficient to control the CO.sub.2 level in each of the rooms and communicate the CO.sub.2 level or a representative control signal to the BMS **142**, such that the CO.sub.2 level in each of the rooms **102** may be maintained by the building management system **142** within a desired operating range.

(32) Throughout operation, the controller **118** may continue to inspect the room air samples from each of the rooms **102** by controlling the air flow controller **106** to cycle each of the plurality of rooms **102** as the selected room **108**. For example, after the controller **118** has received the air quality parameter or measurement from the air quality sensor **140** for a first room (e.g. room 12), the controller **118** may output a signal configured to control the air flow controller **106** to deliver a room air sample from a second room (e.g. room 22) to the air quality sensor **140**. The room air sample for the second room (e.g. room 22) may be measured by the air quality sensor **140**, which may communicate an air quality parameter to the controller **118**. Based on the air quality parameter

from the second room (e.g. room 22), the controller **118** may communicate a ventilation control signal to the BMS **142**. In response to the ventilation control signal, the BMS **142** may control a ventilation, heating, and/or cooling property of the air supplied to the second room (e.g. room 22). In this way, the air sampling device **104** may provide control information and/or measurement information configured to identify an air quality parameter of the each of the plurality of rooms **102**.

(33) Though discussed in reference to the CO.sub.2 sensor, the air quality sensor **140** may correspond to any form of device configured to measure a condition or an air quality parameter of an air sample. For example, the air quality sensor **140** may correspond to one or more forms of volatile organic compound (VOC) sensors, humidity sensors, CO.sub.2 sensors, carbon monoxide (CO) sensors, ozone sensors, etc. Accordingly, the air sampling device **104** may be configured to provide various forms of air quality information to the BMS **142**. In various embodiments, the controller **118** of the air sampling device **104** may be configured to provide various signals, measurements, and/or control instructions to the building management system **142** to suit a particular application of the air sampling system **100**.

(34) The at least one sensor **110** of the system **100** may further comprise an operation sensor **150**, which may correspond to an air flow rate sensor, differential pressure transmitter, pressure sensor, etc. The operation sensor **150** may be connected to an inlet line **152** and an outlet line **154** of the air quality sensor **140**. In this configuration, the operation sensor **150** may be configured to identify if the air flow provided to the air quality sensor **140** is sufficient to measure the air quality parameter of the selected room **108**. Additionally, the operation sensor **150** may be configured to identify a change in pressure over the operating life or maintenance cycle of the air sampling system **100**. For example, the controller **118** may monitor pressure data supplied by the operation sensor **150** to identify a change in pressure over time or a pressure below a predetermined threshold. Such changes in operating pressure may be utilized by the controller **118** to identify a maintenance condition or failure in operation of the air sampling device **104**.

(35) Still referring to FIG. 3, the sample return unit **114** may correspond to an air transport motivator. The sample return unit **114** may correspond to an energy efficient air transport unit configured to function without a need for direct electrical power or a dedicated mechanical drive, vacuum or propulsion system. In such embodiments, the sample return unit **114** may be configured to generate suction to draw or transfer the room air samples **132** from each of the plurality of rooms **102** in response to being positioned within an air flow of a bulk supply fan **160** or air displacement device of the HVAC system **116**. For example, the sample return unit **114** may correspond to a Venturi vacuum device configured to utilize differential pressure from the air flow in the form of a vacuum pump **162** derived from the bulk supply fan **160**. The HVAC system **116** may be configured to provide heating, ventilation, and/or air conditioning to each of the rooms **102**. In this configuration, the air sampling system **100** may be configured to function without a dedicated or air displacement device requiring energy that is not already utilized by operating the HVAC system **116**.

(36) Though the sample return unit **114** is described in the exemplary embodiment as a Venturi vacuum device, in some embodiments, the sample return unit **114** may recover the room air samples utilizing conventional vacuum pumps or fans. For example, in some embodiments, the sample return unit **114** may utilize suction created via a reciprocating pump, screw, or turbine compressor, or various devices configured to displace air in response to receiving electrical current. Such devices may be utilized to provide vacuum pressure to return the room air samples. However, these devices may be noisy, inefficient, and require more maintenance than the Venturi vacuum device disclosed herein.

(37) As discussed herein in the exemplary embodiment, the only energy usage of the sample return unit **114** may be a small parasitic energy drawn from the bulk supply fan **160** of the HVAC system **116**. For example, the bulk supply fan **160** may consume a small and potentially negligible increase

in power to provide the suction necessary to operate the sample return unit **114**. In this way, the air sampling system **100** may improve efficiency while limiting components and related maintenance that may be required when utilizing active devices to provide vacuum pressure to return the room air samples. Instead, the sample return unit **114** may utilize the existing fluid pressure from the HVAC system **116** to generate suction to power the sample return unit **114** via the Venturi effect to improve efficiency and limit maintenance.

(38) In operation, the sample return unit **114** may draw room air samples from each of the plurality of rooms **102** perpetually during operation of the bulk supply fan **160** of the HVAC system **116**. For example, the controller **118** of the air sampling system **100** may be configured to control the air flow controller **106** to direct a room air sample of a selected room **108** (e.g. room 12) to the air quality sensor **140**. The air quality sensor **140** may then measure a condition or quality parameter of the selected room **108**. Additionally, the air flow controller **106** may be configured to retrieve and purge room air samples **132** from each of the idle rooms **112** (e.g. rooms 11, 13, 14, and 21-24). By continuously drawing the room air samples **132** from each of the idle rooms **112** and the selected room **108**, the air sampling system **100** may supply updated air samples to the air flow controller **106** without significant delay due to purging air samples. Accordingly, the air supplied to the at least one sensor **110** may only need to be purged of the air in the sensor supply line **120b** when changing the selected room **108** among the plurality of rooms **102**. In this way, the system may provide for economical monitoring of the air quality in each of the rooms **102** while maintaining a low cycle time for monitoring each of the rooms **102**. Accordingly, the disclosure provides for a responsive monitoring system operable to quickly and effectively provide control information to the BMS **142** to ensure that room air quality is maintained throughout a building while limiting associated cost and maintenance fees.

(39) Referring to FIG. 4, a detailed schematic diagram sample return unit **114** of the air sampling device **104** is shown. As previously discussed, the sample return unit **114** may comprise the vacuum pump **162** configured to derive suction to draw or transfer the air samples from each of the plurality of rooms **102** to the air flow controller **106**. The sample return unit **114** or air supply unit of the system **100** may be disposed in an air flow generated by an air supply unit or the bulk fan supply **160** of the HVAC system **116**. The sample return unit **114** comprises an air inlet **172** and an air outlet **174**. The air inlet **172** is configured to receive an inlet portion **176a** of the air flow **176** and an outlet portion **176b** released through the air outlet **174**.

(40) The inlet portion **176a** of the air flow **176** may have an inlet pressure and an inlet velocity. The outlet portion **176b** of the air flow **176** may have an outlet pressure and an outlet velocity. The sample return unit **114** may generate the suction of a suction line **178** by utilizing a pressure differential between the inlet pressure and a vacuum pressure in a vacuum section **182**. For example, the pressure differential may be generated by the sample return unit **114** due to the Venturi effect causing the inlet pressure to be greater than a vacuum pressure in the vacuum section **182** between the inlet portion **176a** and the outlet portion **176b**. Additionally, the Venturi effect may cause the inlet velocity to be lower than an intermediate velocity between the inlet portion **176a** and the outlet portion **176b**. In this configuration, the difference between the inlet pressure and the vacuum pressure created by the sample return unit **114** generates the suction that draws the room air samples via the vacuum pump **162** between the air inlet **172** and the air outlet **174**.

(41) As previously discussed, the vacuum section **182** is disposed between the air inlet **172** and the air outlet **174**. The vacuum section **182** forms a central cross-section perpendicular to the air flow **176**. The central cross-section is smaller than an inlet cross-section of the air inlet **172** and an outlet cross-section of the air outlet **174**. In this configuration, the changes in the cross-sections among the air inlet **172**, the vacuum section **182**, and the air outlet **174** may result in vacuum pressure to be generated at the connection between the suction line **178** and the vacuum section **182**. In this configuration, the sample return unit **114** may draw the air samples from each of the plurality of rooms **102** to the air flow controller **106**. Accordingly, the air samples may be drawn from the

plurality of rooms **102** throughout the operation of the air sampling system **100** in response to the activation of the air flow generated by the bulk fan supply **160** of the HVAC system **116**.

(42) Referring now to FIGS. 5, 6, and 7; representative diagrams of the remote test environment **122** formed by the sealed enclosure **124** are shown demonstrating the operation of the remote test environment in coordination with the sampling device **104**. As previously discussed, the remote test environment **122** is formed within the sealed enclosure **124**, which may be housed within an enclosure of the sampling device **104**. In operation, the air flow controller **106** may continuously cycle sampled air from each of the rooms **102** by drawing air from the rooms via the suction line **178**. In order to test the selected room **108**, the controller **106** may direct the air sample from the selected room **108** into the sealed enclosure **124**, such that the micro-environment **126** is generated within the interior cavity **128** of the sealed enclosure **124**. In this way, the system **100** may provide for a simulated environment with representative air quality conditions to that of the selected room **108** and each of the rooms **102** connected to the sampling device in a remote location separate from the rooms **102**.

(43) Within the remote test environment **122** formed by the sealed enclosure **124**, at least one of the sensors **110** may be housed, such that the air flow controlled by the air flow controller **106** may be delivered from the selected room **108** into the interior cavity **128**. In this way, the system may generate similar or representative ambient air conditions to those of the selected room **108** within the sealed enclosure **124**, such that the one or more sensors **110** disposed therein may be operable to sample and test the micro-environment **126** and thereby simulate a test of the ambient air conditions within the selected room **108**. The operation of the remote test environment **122** may be cycled for each of the rooms **102** in connection with the sampling device **104** without appreciable lag or delay as a result of the continuous delivery of the room air samples from each of the rooms **102**. In this way, the system **100** may provide for advanced testing of simulated ambient air conditions in each of the rooms **102** while only requiring a limited vacuum pressure and corresponding flow rate of the room air samples supplied to the sampling device **104** by the flow controller **106**.

(44) As discussed herein each of the sensors **110** may correspond to an air quality or particulate air sensor, such as the air quality sensors **140** previously discussed herein. The air quality sensor **140** may correspond to any form of device configured to measure a condition or an air quality parameter of an air sample. For example, the air quality sensor **140** may correspond to one or more forms of volatile organic compound (VOC) sensors, humidity sensors, CO.sub.2 sensors, carbon monoxide (CO) sensors, ozone sensors, flow rate sensors or flow meters, etc. Additionally, the air quality sensors may correspond to electro-chemical sensors that may be configured to detect one or more allergens or particulate materials suspended in the air within the selected room **108**. Accordingly, the air sampling device **104** may be configured to provide various forms of air quality information to the BMS **142**.

(45) The sealed enclosure **124** of the remote test environment may be configured to receive the room air samples from the selected rooms **108** via an enclosure inlet **190**. The air from the selected room **108** forming the micro-environment **126** may further be purged from the interior cavity **128** via the suction line **178**. Inlet air supply line **194** may be connected to the enclosure inlet **190** via a sealed coupling **198**. Similarly, an outlet purge line **196** may be coupled to enclosure outlet **192** via the sealed coupling **198**. In this configuration, the air supply to the micro-environment **126** from the selected room **108** may be enclosed within a sealed boundary wall **200**. In this configuration, the sample air within the remote test environment **122** may form the micro-environment **126** representative of the ambient air quality conditions within the selected room **108**. The sealed boundary wall **200** may prevent seepage from a local environment **202** (e.g. the local environment of the sampling device **104**) from entering the sealed enclosure **124** and interrupting or otherwise limiting accuracy of the representation of the air conditions within the micro-environment **126** relative to those in the selected room **108**.

(46) In operation, the sealed enclosure **124** and boundary wall **200** prevent contamination or seepage from the local environment from entering the interior cavity **128**, which may otherwise adversely impact the accuracy of the test data captured by at least one sensor **110** disposed therein. Preventing seepage from the local environment may be of particular importance for the system **100** in implementations that utilize the sample return unit **114** in the form of a Venturi vacuum device for similarly low energy systems, which may derive differential pressure from air flow controllers for the HVAC system **116** of the building. For example, in systems that utilize sample return units **114** that are only operable to generate limited differential pressure (e.g. less than 2 inches of mercury or 70 mbar), seepage from the local environment **202** may form a significant portion of the air and related contaminants entering the micro-environment **126**. Accordingly, in order to direct sufficient air from the selected room **108** into the micro-environment **126** via the suction line **178**, the interior cavity **128** of the sealed enclosure **124** should be effectively sealed, particularly in systems that are optimized for efficient operation with limited flow rates passing through the interior cavity **128**.

(47) In some implementations, in order to insure the integrity of the sealed boundary wall **200** forming the interior cavity **128**, the system **100** may further comprise a differential pressure sensor **204**. The differential pressure sensor **204** may be in connection with or otherwise configured to detect the pressure between the enclosure inlet **190** and the enclosure outlet **192** in order to insure that the sealed boundary wall **200** has not been breached or otherwise compromised. Accordingly, the differential pressure sensor **204** may be in communication with the controller **118**, such that the controller **118** may monitor pressure across the micro-environment **126** and thereby test the integrity of the sealed enclosure **124** and the sealed boundary wall **200**. As demonstrated, the pressure sensor **204** is configured to detect an external pressure differential in order to ensure that the integrity of the boundary wall **200** and the connections (e.g., the inlet **190**, outlet **192**), and any external fittings are maintained such that the environment within the seal enclosure **124** is isolated from the environment outside the boundary wall **200**. In this way, the system may diagnose potential leaks and various system operating defects in response to changes detected in the differential pressure.

(48) As depicted in FIG. 5, the at least one sensor **110** may comprise a first sensor apparatus **110a**, a second sensor apparatus **110b**, and/or a third sensor apparatus **110c**. Though three sensors are discussed in the exemplary implementation demonstrated in FIG. 5, it may be understood that additional sensors may be implemented without departing from the spirit of the disclosure. One or more of the sensors **110** may comprise an air transfer unit **210**. Air transfer unit **210** may correspond to a fan, pump, or other air transfer devices that may be configured to electively communicate air within the micro-environment **126** and a housing or enclosure **212** of the one or more sensors **110**. For example, the first sensor apparatus **110a** may comprise a first air transfer unit **210a**, and the second sensor apparatus **110b** may comprise a second air transfer unit **210b**. In this configuration, the controller **118** may control each of the first sensor apparatus **110a** and the second sensor apparatus **110b** to independently control the flow of the ambient air within the micro-environment **126** through their respective enclosures **212** in order to optimize operation of each of the sensors. One or more of the sensors **110**, in the third exemplary sensor apparatus **110c**, may be configured to operate in the localized conditions of the air within the micro-environment **126** without requiring a separate air transfer unit **210**. Accordingly, the remote test environment **122** formed by the sealed enclosure **124** may provide for a remote test apparatus **220** that may be flexibly implemented to simulate and detect the ambient air conditions in each of the rooms **102**.

(49) In order to avoid interference among the sensors **110** in the micro-environment, the arrangement of the sensors **110** may be provided in a sequential order or along unique or otherwise non-overlapping flow paths within the micro-environment **126**. For example, in the configuration shown in FIG. 5, the first sensor **110a** may be a temperature sensor, the second sensor **110b** may be a particulate air sensor, and the third sensor **110c** may be a flowrate sensor or flow meter. More

generally, in cases where a temperature sensor is included in the micro-environment, the temperature sensor may be arranged upstream from any sensors, particularly those that generate heat or otherwise could adjust a temperature of air supplied to the temperature sensor along the flow path identified by the arrows in inlet air supply line **194** and the suction line **178**. In this arrangement, heat that may be introduced by the second or third sensors **110b**, **110c** is downstream of the air flow from the first sensor **110a**, which preserves the accuracy of temperature readings by the first sensor **110a**.

(50) In addition to the first sensor **110a** implemented as the temperature sensor positioned proximate to the inlet **190**, the flow rate sensor may be positioned proximate to the outlet **192** similar to the third sensor **110c**. Additionally, as discussed later in reference to FIG. 7, the first sensor **110a**, the second sensor **110b**, and/or the third sensor **110c** may be arranged such that the flow of air within the micro-environment **126** does not result in the air interacting with the first sensor **110a** being received by the second or third sensors **11b**, **110c**. More generally, the sensors in the micro-environment **126** or the sealed enclosure **124** may be arranged such that the directional flow of the air received at the inlet **190** and removed via the outlet **192** does not sequentially flow from one sensor **110** to another sensor **110** in cases where the sensors implemented may introduce heat, disrupt flow, or introduce other variables within the enclosure **124** that may affect the accuracy and response of the data collected by the sensors. Stated simply, the sensors **100** may be arranged such that fresh air samples are delivered to those sensors that are sensitive to variables that would be introduced in the wake of an upstream sensor.

(51) Referring now to FIGS. 5 and 6, the first sensor apparatus **110a** and the second sensor apparatus **110b** are demonstrated in a first flow rate **222** generated by the differential pressure of the suction line **178** generated by the sample return unit **114**. The first flow rate **222** may be drawn through the interior cavity **128** of the sealed enclosure **124** in a first direction **224**, which may be substantially between the enclosure inlet **190** and the enclosure outlet **192**. As previously discussed, each of the first sensor apparatus **110a** and the second sensor apparatus **110b** may be disposed within the micro-environment **126** formed by the enclosure **124**, such that the sensors **110a**, **110b** may receive the air passing through the micro-environment **126** provided from the selected room **108** at the first flow rate **222**. In this configuration, each of the sensors **110a**, **110b** may be configured to independently sample the simulated conditions from the selected room **108** within the micro-environment **126** by controlling their respective air transfer units **210a**, **210b**.

(52) First sensor apparatus **110a** may comprise the first air transfer unit **210a**. The first air transfer unit **210a** may be configured to draw air through the enclosure **212** at a second flow rate **226**. Additionally, first air transfer unit **210a** and enclosure **212** of the first sensor apparatus **110a** may be oriented within the interior cavity **128**, such that second flow rate **226** is directed in a second direction **228**, which may be substantially aligned with the first direction **224** of the first flow rate **222**. The second air transfer unit **210b** within the enclosure **212** of the second transfer apparatus **110b** is configured to draw or transfer the air from the micro-environment **126** through the enclosure **212** at a third flow rate **230**. As depicted in FIG. 6, the air flow communicated by the second transfer unit **210b** is directed in a third direction **232**. The third direction **232** may be substantially opposite the first direction **224** of the flow controlled by the sample return unit **114**. Accordingly, flow rate and direction of the sensors **110** may be independently controlled by the system **100** in order to provide for flexibility in operation.

(53) As discussed herein, the directions **224**, **228**, and **232** of the respective flow rates **222**, **226**, and **230** are described as being substantially aligned or in substantial opposing directions. However, the directions **224**, **228**, and **232** of each of the flow rates **222**, **226**, and **230** may be arranged perpendicular, skewed, or in a variety of orientations relative to first direction **224** without departing from the spirit of the disclosure. As described herein, the term “substantially” in reference to the directions provides for variations in manufacturing, assembly, and time constraints of the interior cavity **128** micro-environment **126** that may result in alterations in the direction of

the flow rates **222**, **226**, and **230** to various degrees. Accordingly, the term “substantially” may provide for variations of ± 10 degrees of alignment, which may provide for similar operating results while still maintaining the general relative directions described herein. Finally, while discussed in reference to the flow rates **222**, **226**, and **230**, additional sensors **110** may include additional air transfer units **210** or may combine air transfer units in a variety of ways to independently control any number of flow rates for corresponding sensors that may be implemented within the constraints of the interior cavity **128**.

(54) Referring now to FIGS. **6** and **7**, the interior cavity **128** of the sealed enclosure **124** may comprise one or more contoured walls **240** formed by partitions **242** of sealed enclosure **124** or may be formed from the same mold as the sealed enclosure **124**. In general, the contoured interior walls **240** may provide for contoured shape of the micro-environment **126** approximate the enclosure inlet **190** and enclosure outlet **192**. In this configuration, sealed enclosure **124** may provide for an expanded central region **244** enclosed between the enclosure inlet **190** and the enclosure outlet **192**. The expanded central region **244** may provide for a volume of air from the selected room **108** to be housed within the enclosure **124**, such that the one or more sensors **110** enclosed within interior cavity **128** have ample air within the micro-environment **126** to effectively simulate the selected room **108**. The proportions of the interior cavity **128** and the expanded central region **244** may be dependent upon the particular type of operating environments for each of the sensors **110**. Accordingly, the proportions of the micro-environment **126** may be adjusted or varied to suit a variety of applications based on the teachings of the disclosure.

(55) As depicted in FIG. **6**, contoured interior walls **240** are demonstrated in reference to a side cross-sectional view of the cavity **128**. As shown, the sensors **110** may be disposed along a base **246** of the sealed enclosure **124** which may extend proximate to enclosure inlet **190** and enclosure outlet **192**. In this configuration, the air supply lines **194**, **196** may pass through the side walls **248** of the sealed enclosure **124**, such that the enclosure inlet **190** and enclosure outlet **192** are proximate to the base **246**. In this configuration, the first flow rate **222** flows through the interior cavity **128** via the suction line **178** may be aligned with the sensors **110** disposed along the base **246** of the sealed enclosure **124** within the interior cavity **128**.

(56) As demonstrated in FIG. **7**, a top cross-sectional view of the interior cavity **128** demonstrates walls **240** opening to the expanded central region **244**, enclosure inlet **190** and enclosure outlet **192**. As shown, the sensors **110** range such that air communicated from the selected room **108** to the suction line **178** is directed across the enclosures **212** of each of the sensors **110**. Additionally, the first sensor apparatus **110a** and the second sensor apparatus **110b** are arranged along the contoured interior walls **240**, while the third sensor apparatus **110c** is disposed centrally within the expanded central region **244** formed by the interior cavity **128**. The arrangement demonstrated in FIG. **7** may be beneficial in that the transfer units **210** of the first sensor apparatus **110a** and the second sensor apparatus **110b** may communicate the air through the sensor enclosures **212**, such that their position in the interior cavity **128** is less critical than that of the third sensor apparatus **110c**, which is depicted without the air transfer unit **210**. Such an arrangement may be beneficial in circumstances where the specific proportions or limitations of the first flow rate create limited flow zones **250** along the contoured interior walls **240**. Accordingly, first and second air transfer units **210a**, **210b** may circulate the air through the enclosures **212** of the first sensor apparatus **110a** and the second sensor apparatus **110b** to improve the circulation of the air through the limited flow zones formed within the interior cavity **128**. In this way, the operation of the system **100** may be optimized based on the particular sensors implemented therein.

(57) As previously discussed, one or more of the sensors **100** may be arranged within the micro-environment **126** such that fresh air samples are delivered to those sensors that are sensitive to variables that would be introduced in the wake of an upstream sensor. As demonstrated in FIG. **7**, the first sensor **110a**, the second sensor **110b**, and/or the third sensor **110c** may be arranged such that the flow of air within the micro-environment **126** does not result in the air interacting with the first

sensor **110a** being received by the second or third sensors **110b**, **110c**. This may be particularly important in cases where the first sensor **110a** is a temperature sensor could be detect heat emitted from an upstream sensor if not arranged to receive air freshly supplied from the inlet **190**. More generally, the sensors in the micro-environment **126** or the sealed enclosure **124** may be arranged such that the directional flow of the air received at the inlet **190** and removed via the outlet **192** does not sequentially flow from one sensor **110** to another sensor **110** in cases where the sensors implemented may introduced heat, disrupt flow, or introduce other variables within the enclosure **124** that may affect the accuracy and response of the data collected by the sensors.

(58) Referring now to FIG. **8**, a flow chart is shown demonstrating an exemplary operation of the system **100** comprising the remote test apparatus **220**. The method **260** may begin by initiating operation for rooms **102** of the building **10**. The operation may be initiated by the controller **118** controlling the sample return unit **114** to select the air sample from the selected room **108** to be delivered to the micro-environment **126** (**262**). Once initiated, the air supplied to the micro-environment **126** from the selected room **108** may be purged from the interior cavity **128** out through the suction line **178** for a predetermined time period (**264**). Once the air within the micro-environment **126** is purged over the predetermined time period, the air within the interior cavity **128** is representative of the ambient air conditions of the selected room **108**. Once the micro-environment **126** is representative of the selected room, the method **260** may continue by initiating detection routines for one or more sensors **110** (**266**).

(59) With reference to the first sensor apparatus **110a**, the controller **118** may activate the first air transfer unit **210a** to circulate the air from the micro-environment **126** through the enclosure **212** at the first flow or circulation rate (**268**). Following the activation of the first air transfer unit **210a**, the controller may command the sensor within the first sensor apparatus **110a** to detect a first or multiple characteristics of the air sample from the micro-environment **126** (**270**). Once the results are captured by the first sensor apparatus **110a**, controller **118** may monitor and receive results from the first sensor apparatus **110a** and report them or otherwise utilize the results to report the air quality and/or control the building management system **142** (**272**). Following step **272**, the controller **118** may control the sampling device **104** to select the next room in sequence as the selected room **108** of the rooms **102** and return to step **262** (**274**).

(60) In reference to the second sensor apparatus **110b**, controller **118** may activate the second air transfer unit **210b** to circulate the air from the micro-environment **126** through the enclosure **212** at the second flow or circulation rate (**278**). Following the activation of the second air transfer unit **210b**, the controller may command the sensor within the second sensor apparatus **110b** to detect a first or multiple characteristics of the air sample from the micro-environment **126** (**280**). Once the results are captured by the second sensor apparatus **110b**, controller **118** may monitor and receive results from the second sensor apparatus **110b** and report them or otherwise utilize the results to report the air quality and/or control the BMS **142** (**282**). Following step **282**, the controller **118** may control the sampling device **104** to select the next room in sequence as the selected room **108** of the rooms **102** and return to step **262** (**274**).

(61) Similar to the first sensor apparatus **110a** and the second sensor apparatus **110b**, the controller may activate the third sensor apparatus **110c** to detect one or more quality factors or metrics related to forming the micro-environment **126** from the selected room **108** (**286**). The results from the third sensor apparatus **110c** may be communicated to the controller **118** in accordance with the disclosure (**288**). Following step **284**, the method **260** may return to step **274** as previously discussed. Accordingly, the system **100** may be configured to simultaneously control the operation of the first sensor apparatus **110a**, the second sensor apparatus **110b**, and the third sensor apparatus **110c** to monitor the micro-environment **126**. In this way, the controller **118** may detect the quality metrics identified by each of the sensors **110** in parallel or over a combined, overlapping temporal period within the single micro-environment **126**.

(62) Additionally, as the conditions within the micro-environment **126** are representative of the

ambient conditions within the selected room **108**, the controller **118** may apply a comparative analysis to the information communicated from one or more of the sensors **110**. The comparison of the results by the controller **118** from the sensors **110** may be utilized to identify a fault or cross-reference results between or among the quality metrics identified by the sensors **110**. In this configuration, the micro-environment **126** and corresponding sensors **110** may provide comparative information for the conditions within the micro-environment **126**, which may be utilized for troubleshooting, sensor redundancy, and/or determinations of the validity of the results communicated from the sensors **110**. Accordingly, the utilization of the micro-environment **126** may provide for various benefits to improve the reliability, performance, and capability of the system **100**.

(63) It will be appreciated that embodiments of the disclosure described herein may be comprised of one or more conventional processors and unique stored program instructions that control one or more processors to implement, in conjunction with certain non-processor circuits, some, most, or all of the functions of an image sensor system and method thereof, as described herein. The non-processor circuits may include, but are not limited to, signal drivers, clock circuits, power source circuits, and/or user input devices. Alternatively, some or all functions could be implemented by a state machine that has no stored program instructions, or in one or more application specific integrated circuits (ASICs), in which each function or some combinations of the functions are implemented as custom logic. Of course, a combination of the two approaches could be used. Thus, the methods and means for these functions have been described herein. Further, it is expected that one of ordinary skill, notwithstanding possibly significant effort and many design choices motivated by, for example, available time, current technology, and economic considerations, when guided by the concepts and principles disclosed herein will be readily capable of generating such software instructions and programs and ICs with minimal experimentation.

(64) It should be appreciated by those skilled in the art that the above described components may be combined in additional or alternative ways not explicitly described herein. Modifications of the various implementations of the disclosure will occur to those skilled in the art and to those who apply the teachings of the disclosure. Therefore, it is understood that the embodiments shown in the drawings and described above are merely for illustrative purposes and not intended to limit the scope of the disclosure, which is defined by the following claims as interpreted according to the principles of patent law, including the Doctrine of Equivalents.

Claims

1. A remote test apparatus configured to monitor an environment of at least one remotely located room and configured to receive at least one air sample from the at least one remotely located room, the apparatus comprising: a sealed enclosure forming an interior cavity and comprising an inlet in connection with the at least one remotely located room and an outlet in connection with a sample collection unit, wherein the sample collection unit is configured to communicate the at least one air sample from the remotely located room at a first flow rate; at least one sensor disposed in a housing in fluid communication with the interior cavity; and at least one air transfer unit configured to transfer test air from the interior cavity of the sealed enclosure to the housing of the sensor at a second flow rate different from the first flow rate.
2. The remote test apparatus according to claim 1, wherein the at least one sensor comprises a first sensor and a second sensor.
3. The remote test apparatus according to claim 2, wherein the at least one air transfer unit comprises a first air transfer unit configured to transfer the test air from the interior cavity of the sealed enclosure to the first sensor at the second flow rate.
4. The remote test apparatus according to claim 3, wherein the at least one air transfer unit comprises a second air transfer unit configured to transfer the test air from the interior cavity of the

sealed enclosure to the second sensor at the third flow rate.

5. The remote test apparatus according to claim 4, wherein the second flow rate is different than the third flow rate.

6. The remote test apparatus according to claim 4, further comprising: a controller, wherein the controller is configured to detect a first air quality condition via the first sensor and a second air quality condition via the second sensor.

7. The remote test apparatus according to claim 6, wherein the controller is configured to independently control a first test cycle time of the first sensor and a second test cycle time of the second sensor over a test period.

8. The remote test apparatus according to claim 7, where the first test cycle time and the second test cycle time overlap within the test period.

9. The remote test apparatus according to claim 1, wherein the at least one sensor is disposed in a sensor enclosure in connection with the at least one air transfer unit, wherein the at least one air transfer unit is configured to supply the test air from the interior cavity to the at least one sensor at the second flow rate, wherein the sample collection unit is configured to communicate the at least one air sample from the remotely located room at the first flow rate in a first direction extending from the inlet to the outlet.

10. The remote test apparatus according to claim 9, wherein the at least one air transfer unit is configured to transfer the test air through the enclosure in a second direction different than the first direction.

11. The remote test apparatus according to claim 1, further comprising: a pressure sensor configured to detect a pressure differential between the interior cavity and a local environment proximate the sealed enclosure; and a controller configured to detect a system error in response to changes in the pressure differential.

12. The remote test apparatus according to claim 1, wherein the housing at least partially partitions the sensor from the interior cavity and the test air is transferred to the sensor in the housing by at least one of a fan or pump.

13. A method for monitoring an environmental condition of at least one remotely located room, the method comprising: supplying at least one air sample at a first flow rate from the at least one remotely located room to an interior cavity of a sealed enclosure; transferring a portion of the at least one air sample from within the interior cavity to a second sensor in communication with interior cavity with an air transfer device at a second flow rate different than the first flow rate; and detecting the environmental condition of the portion of the at least one air sample at the second flow rate, wherein a test environment within the interior cavity generates the environmental condition representative of the remotely located room.

14. The method according to claim 13, wherein the portion of the at least one air sample is actively transferred through an enclosure housing the second sensor.

15. A remote test apparatus configured to monitor an environment of at least one remotely located room and configured to receive at least one air sample from the at least one remotely located room, the apparatus comprising: a sealed enclosure forming an interior cavity formed by a sealed boundary wall, the sealed enclosure comprising an inlet in connection with the at least one remotely located room and an outlet in connection with a sample collection unit, wherein the at least one sample collection unit is configured to communicate the at least one air sample from the remotely located room at a first flow rate; and at least one sensor disposed in a sensor enclosure in the interior cavity; wherein the at least one sample collection unit is configured to supply a portion of the at least one air sample to the sensor enclosure at a second flow rate different from the first flow rate.

16. The remote test apparatus according to claim 15, further comprising: a pressure sensor configured to detect a pressure differential within the sealed enclosure derived from the operation of the sample collection unit; and a controller configured to detect an operational error in response

to changes in the pressure differential, wherein the operational error indicates a leak in the sealed boundary wall allowing air from the local environment to enter the sealed enclosure.

17. The remote test apparatus according to claim 15, wherein the at least one air sample at the first flow rate passes along a flow path directionally extending from the inlet to the outlet and the at least one sensor comprises a first sensor and a second sensor.

18. The remote test apparatus according to claim 17, wherein the first sensor is a temperature sensor and the first sensor is arranged upstream along the flow path from the second sensor.

19. The remote test apparatus according to claim 18, wherein the temperature sensor positioned proximate the inlet and the second sensor generates heat in operation.
